

*Provide the class of concrete shown on Girder Schedule.
Keep girders in an upright position w/the support points within 2'-0" of the girder ends.
Deflected strands may be bundled between hold-down points.
Provide temporary diaphragm beams as required, see BR350.*



The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without first consulting a Registered Professional Engineer.

DATE		REVISION DESCRIPTION	
01-2022		Revised welded wire reinforcement specification,	

CALC. BOOK NO. - - -	N/A - - -	SDR DATE - 05-JAN-2022 -	BR310
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3R321 dan

Note: Provide "a" > 14

* Use in line diaphragm beams when $\theta \geq 65^\circ$. Provide conc. inserts in exterior girder webs and 2" dia. holes thru interior girder webs for 4-#7 x 6'-0" bars.

2-#5 "U6" bars at
each end of girder

1" dia. hole for cable restraint
when Alternate A restraint is used
(typ.) "x" = 0.50 girder depth. —

6" to 2'-6"
equal spacing
¢ dual hold downs
(deflected strands > 14)

g.—

 L

DETENSIONING NOTE
Detension strands in
systematic order above
girder ϕ to limit
eccentricity. (8)

TYPICAL DETENSIONING ORDER

Diagram illustrating a 6-strand braid structure. The strands are numbered 1 through 6, arranged in two columns. The left column contains strands 8, 9, 12 and the right column contains strands 2, 3, 6. Arrows indicate crossings: strand 8 crosses over strand 2, strand 9 crosses over strand 3, and strand 12 crosses over strand 6. Labels include "Girder" pointing to the vertical line between columns, and "Strands" pointing to the two columns of circles.

4'-4"

Slant bars as required
to fit inside "U5" stirrups.

BAR "U"
(ASTM A706)

BAR "U3a"
(Use in lieu of "U3" bars
when deflected strands conflict).

BAR BENDING DETAILS

*Provide the class of concrete shown on Girder Schedule.
Keep girders in an upright position w/the support points within 2'-0" of the girder ends.
Deflected strands may be bundled between hold-down points.
Provide temporary diaphragm beams as required, see BR350.
Ensure girder ends are plumb in their final position under full dead load.*

– $\notin$ holes or inserts

4-1" dia. conc. inserts
(* in exterior girder webs and
4-2" dia. holes thru interior
girder webs for 4-#7 x 6'-0"
bars (unless shown otherwise
on detail plans)

Strand c.g.-

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¢ Bearing →

AHEAD END OF GIRDER
(on stationing)

Upper group deflected strands (10 max.) $\rightarrow$

- ✓ Middle group deflected strands (10 max.)

Lower deflected strand group (10 max.)

0.6" spacing
deflected strands
at midspan

"y_c"
c. g. strand
at midspan

"hc" $\geq 3"$ at midspan

— Chamfer soffit
corners 1" max.

BULB DETAIL

Accompanied by dwgs. BR350 and DET3322

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All materials shall be in accordance with the current Oregon Standard Specifications.

OREGON STANDARD DRAWINGS
BT90 AND BT96 (PRECAST
PRESTRESSED CONCRETE
GIRDERS)

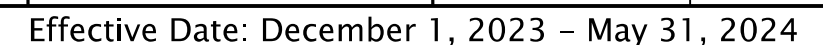
2024

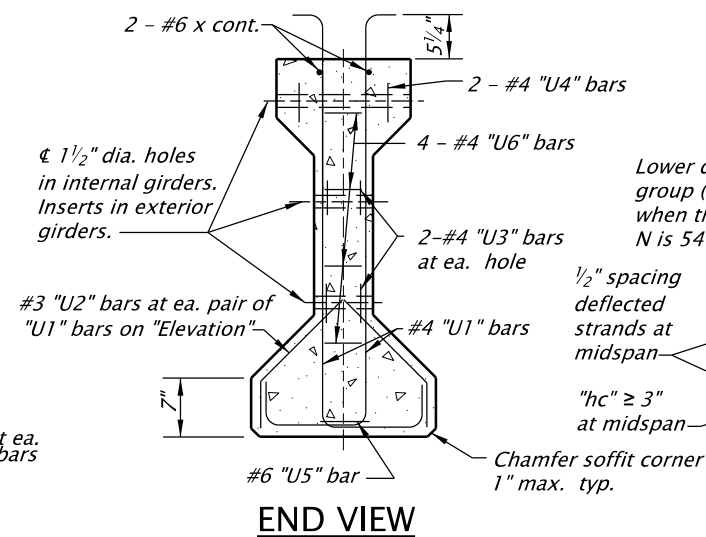
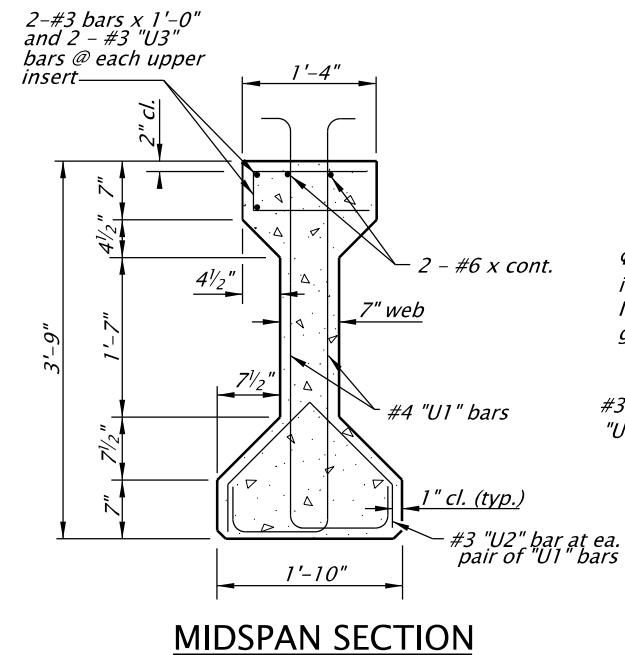
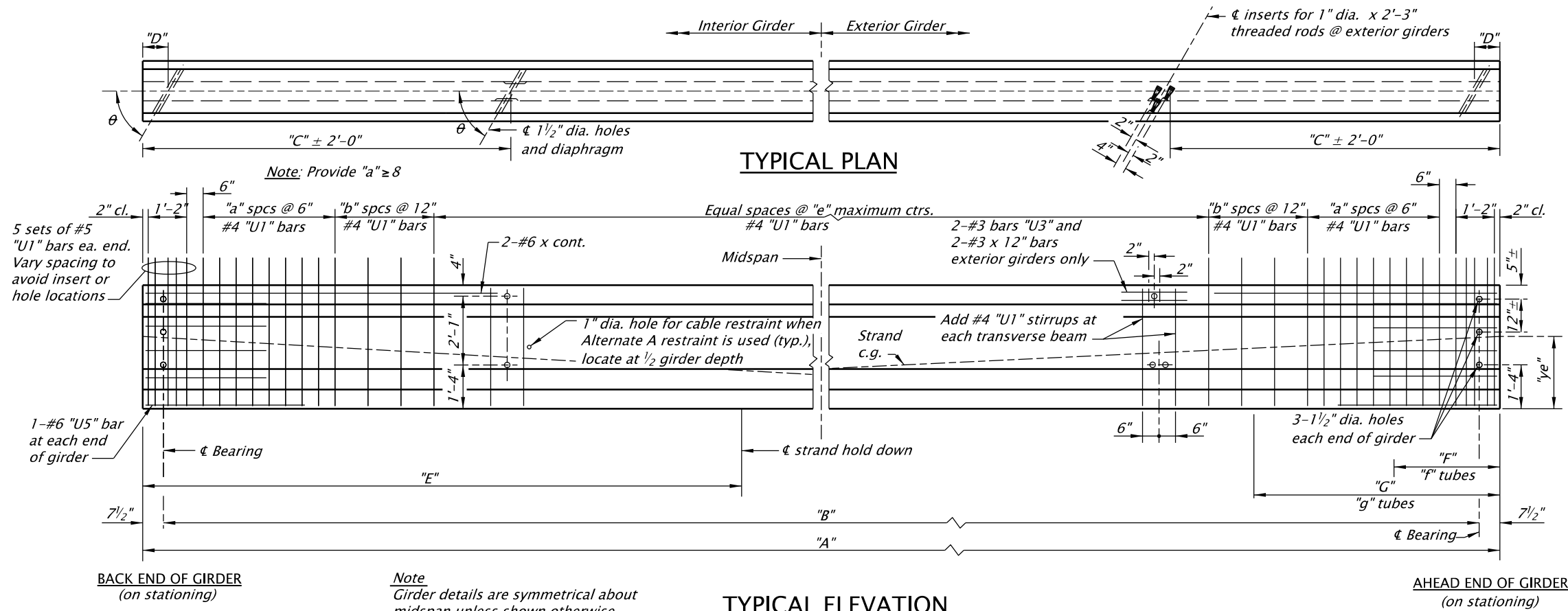
DATE	REVISION DESCRIPTION
01-2022	Revised welded wire reinforcement specification

CALC. BOOK NO. - - - - N/A - - - -	SDR DATE - 05-JAN-2022
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BR321

Effective Date: December 1, 2023 – May 31, 2024



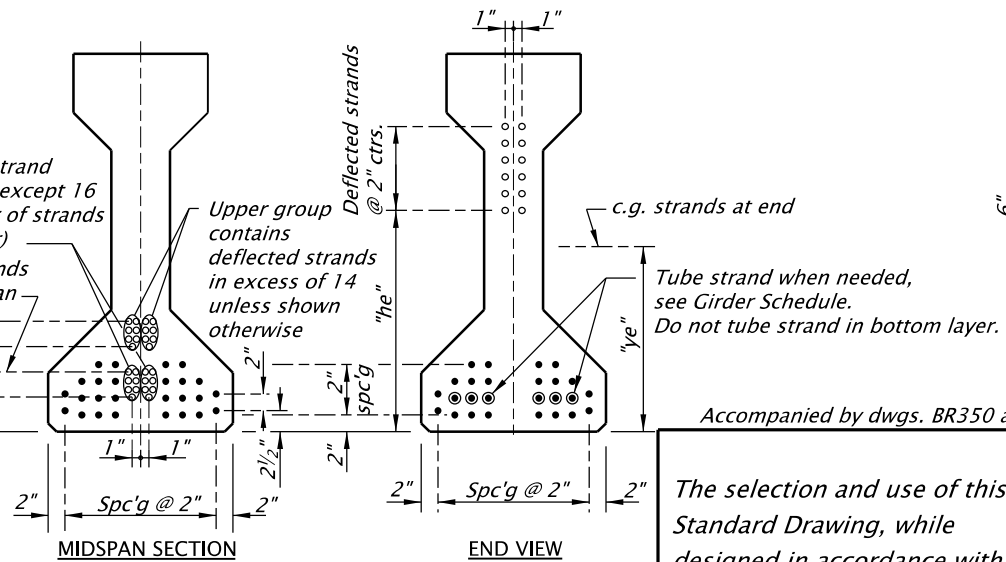


Lower deflected strand group (14 max. except 16 when the number of strands N is 54 or greater)

1/2" spacing deflected strands at midspan

c.g. strands at midspan

"hc" ≥ 3" at midspan



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PRETENSIONING STEEL

Provide 1/2" dia. 7-wire low relaxation strands conforming to AASHTO M203 (ASTMA416), Grade 270, Supplement 1. Provide an initial tension (after harping deflected strands) of 31.0 kips per strand unless shown otherwise. Deflected strands may be bundled between hold down points. Provide the same tension as the straight strands after deflecting. Detension strands alternately about the girder to limit prestress eccentricity.

MILD STEEL REINFORCEMENT

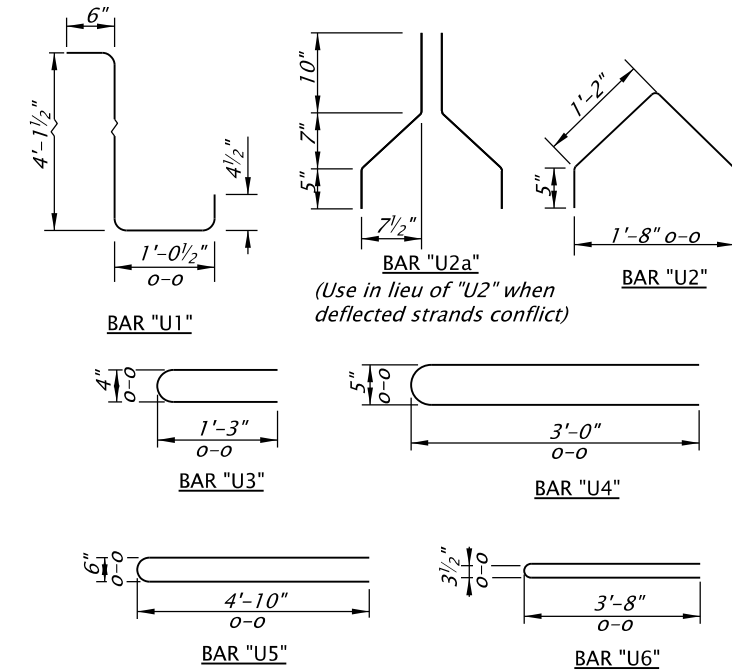
Provide reinforcing steel as specified in the General Notes for the project unless shown otherwise.

HANDLING PRESTRESSED CONCRETE GIRDERS

Keep girders in an upright position with the support points within 2'-0" of the girder ends. Provide temporary diaphragm beams as required, see BR350.

DIAPHRAGM BEAMS

See structure plans for diaphragm beam locations. Pour transverse beams a minimum of three days before deck pour. Provide structural concrete inserts: Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods AASHTO M314, Gr. 36 (ASTM A307) threaded full length. Brace prestressed girders to prevent lateral deflection during diaphragm pour.



Accompanied by dwgs. BR350 and DET3345

All materials shall be in accordance with the current Oregon Standard Specifications.

OREGON STANDARD DRAWINGS

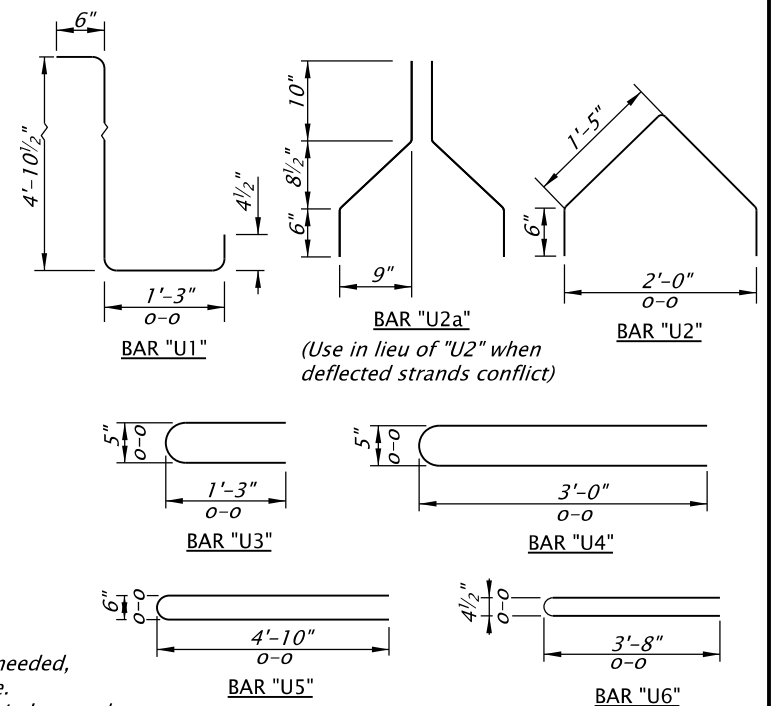
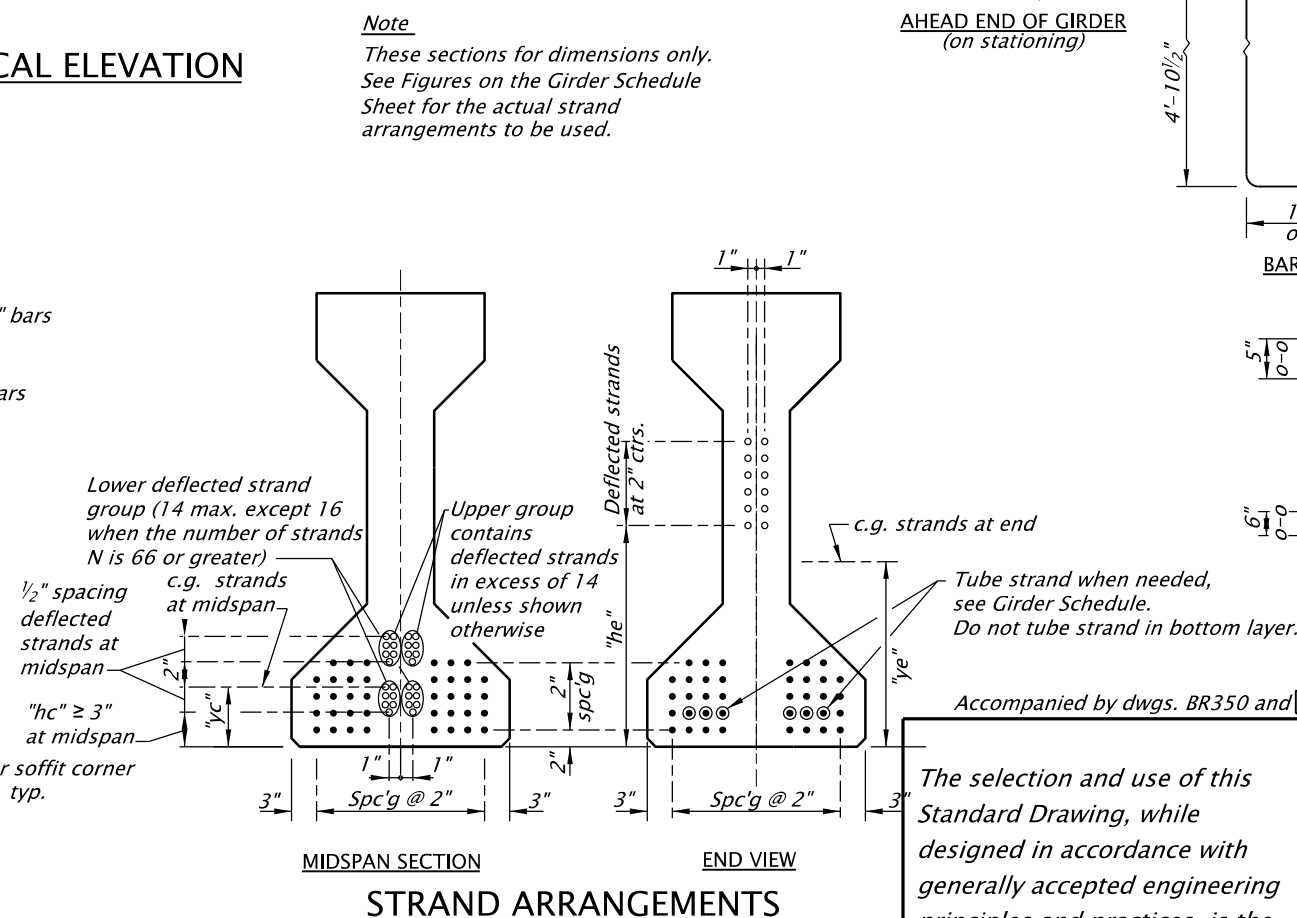
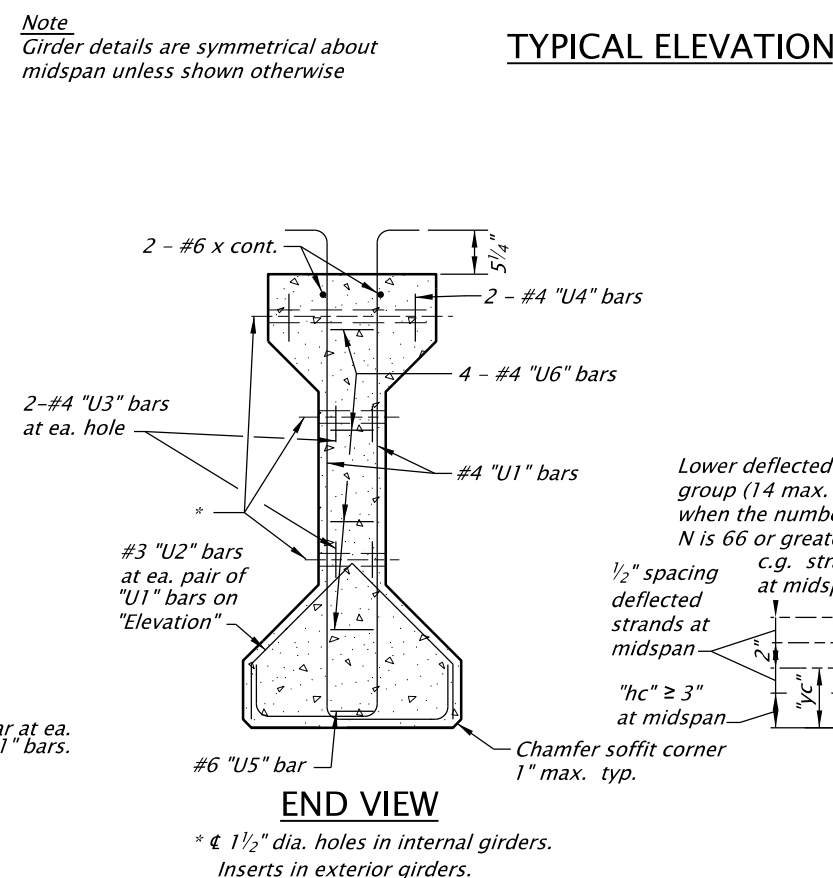
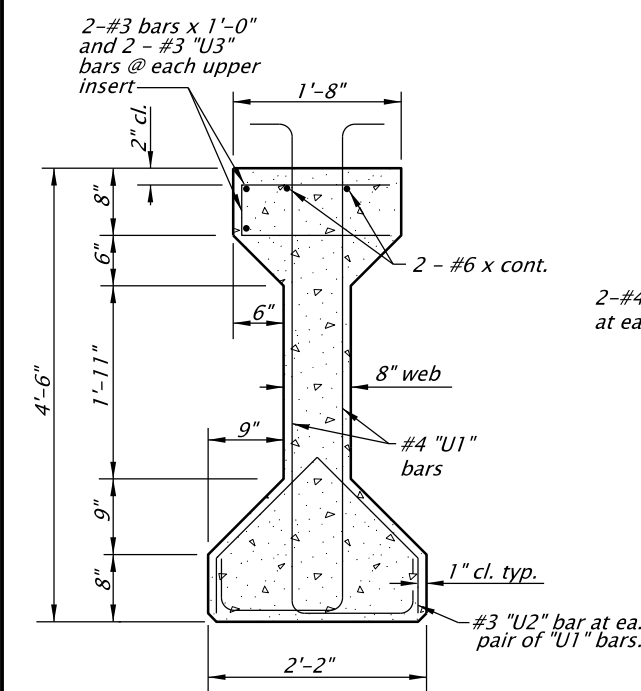
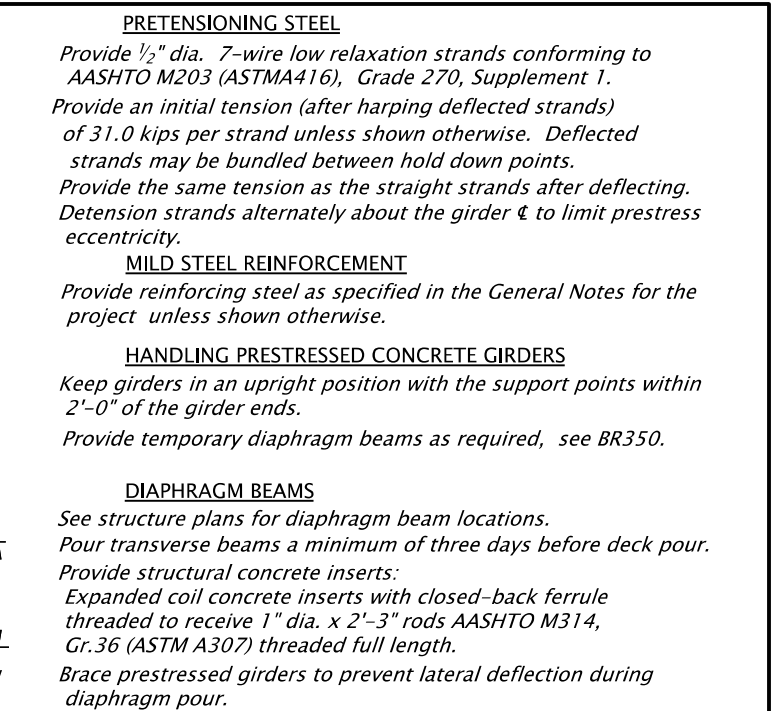
TYPE III - PRESTRESSED CONCRETE GIRDERS

2024

DATE	REVISION	DESCRIPTION
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-	-	-
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CALC. BOOK NO.	N/A	SDR DATE: 20-APR-2018

BR330

Effective Date: December 1, 2023 – May 31, 2024



Accompanied by dwgs. BR350 and DET3345

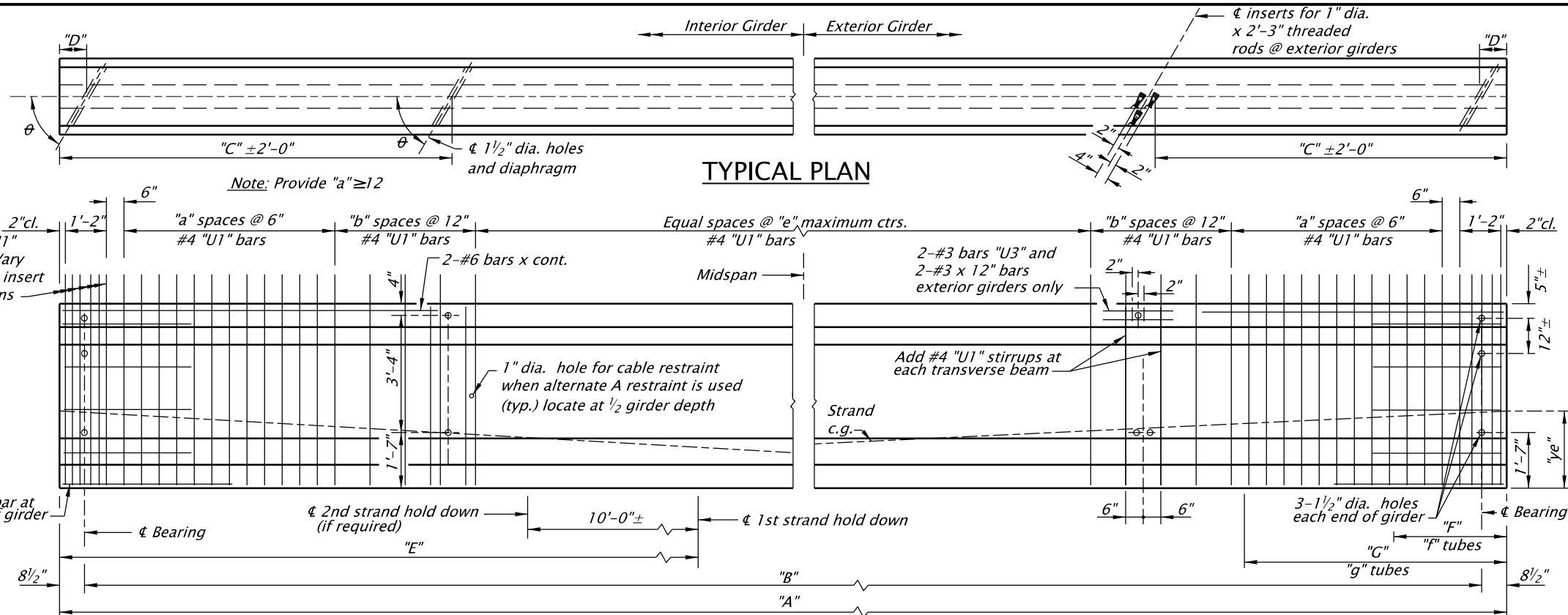
The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without first consulting a Registered Professional Engineer.

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TYPE IV - PRESTRESSED CONCRETE GIRDERS

2024

DATE		REVISION DESCRIPTION	
-	-		
CALC. BOOK NO. - - - - N/A - - - -		SDR DATE 20-APR-2018 -	BR335

**PRETENSIONING STEEL**

Provide 1/2" dia. 7-wire low relaxation strands conforming to AASHTO M203 (ASTMA416), Grade 270, Supplement 1. Provide an initial tension (after harping deflected strands) of 31.0 kips per strand unless shown otherwise. Deflected strands may be bundled between hold down points. Provide the same tension as the straight strands after deflecting. Detension strands alternately about the girder ϵ to limit prestress eccentricity.

MILD STEEL REINFORCEMENT

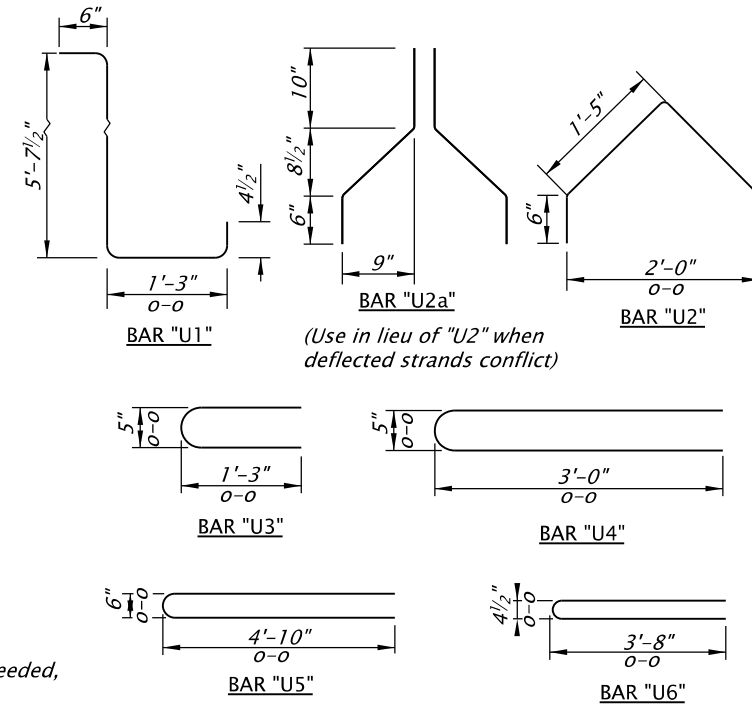
Provide reinforcing steel as specified in the General Notes for the project unless shown otherwise.

HANDLING PRESTRESSED CONCRETE GIRDERS

Keep girders in an upright position with the support points within 2'-0" of the girder ends. Provide temporary diaphragm beams as required, see BR350.

DIAPHRAGM BEAMS

See structure plans for diaphragm beam locations. Pour transverse beams a minimum of three days before deck pour. Provide structural concrete inserts: Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods AASHTO M314, Gr. 36 (ASTM A307) threaded full length. Brace prestressed girders to prevent lateral deflection during diaphragm pour.



Accompanied by dwgs. BR350 and DFT3345

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All materials shall be in accordance with the current Oregon Standard Specifications.

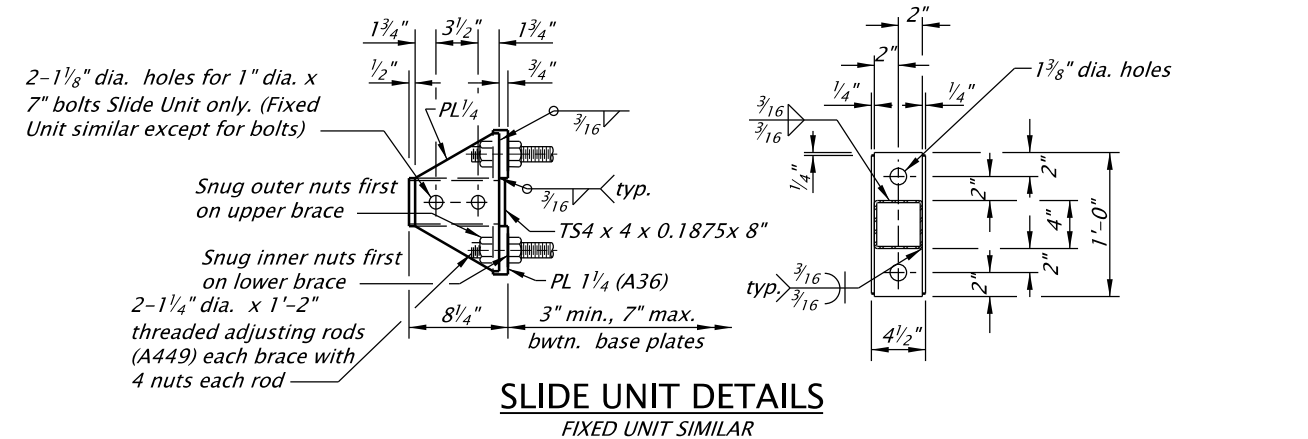
OREGON STANDARD DRAWINGS**TYPE V - PRESTRESSED CONCRETE GIRDERS**

2024

DATE	REVISION	DESCRIPTION
-	-	-
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CALC. BOOK NO. - - -	N/A - - -	SDR DATE- 20-APR-2018 -

BR340

Effective Date: December 1, 2023 – May 31, 2024

[illegible]

BRACE END DETAIL

Install temporary diaphragm beams at exterior girders between end beams and permanent diaphragms prior to pouring permanent diaphragms and end beams. Space temporary diaphragms as needed to resist construction loads but not at more than 25 ft. apart. Assume the deck finishing machine carriage is at the most adverse position when computing construction loads. Remove temporary diaphragms after removing the deck overhang brackets.

Provide pipe conforming to ASTM Specification A53, Grade A or B. Provide structural tubing conforming to ASTM Specification A500, Grade A or B (or ASTM A513 with the strength and elongation characteristics of A500, Grade A or B).

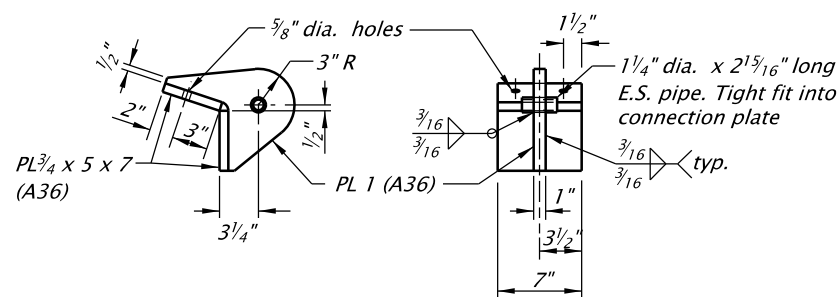
Provide other steel conforming to AASHTO Specification M183 (ASTM A36).

Provide mechanically galvanized high strength bolts conforming to ASTM Specification A325 unless shown otherwise. Provide mechanically galvanized threaded rods conforming to ASTM Specification A449.

Tighten but do not torque high strength bolts and rods.

Hot-dip Galvanize all structural steel after fabrication.

Approved alternate designs prepared by a registered Professional Engineer may be used for temporary diaphragms.



LOWER CONNECTION
PLATE DETAIL
BULB-I and BULB-T GIRDERS

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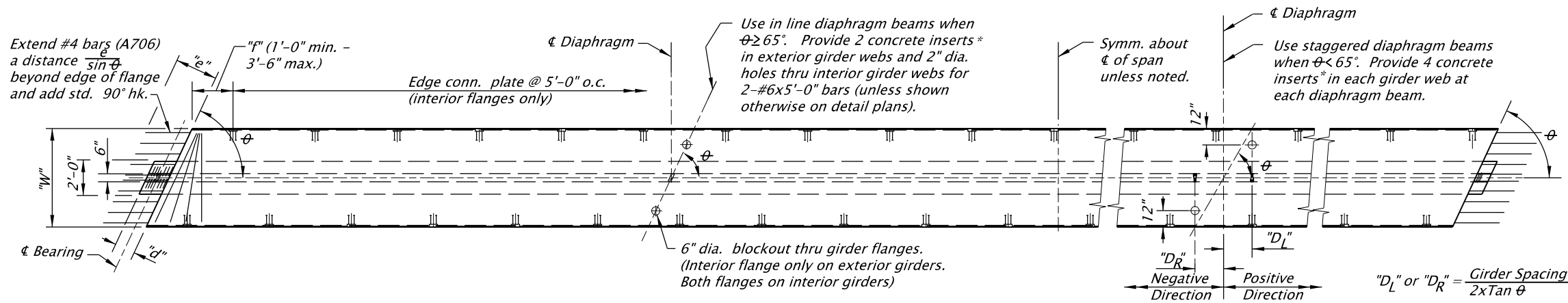
All materials shall be in accordance with the current Oregon Standard Specifications.

OREGON STANDARD DRAWINGS
TEMPORARY DIAPHRAGM
BEAM FOR PRESTRESSED
CONCRETE GIRDERS

2024

DATE		REVISION DESCRIPTION	
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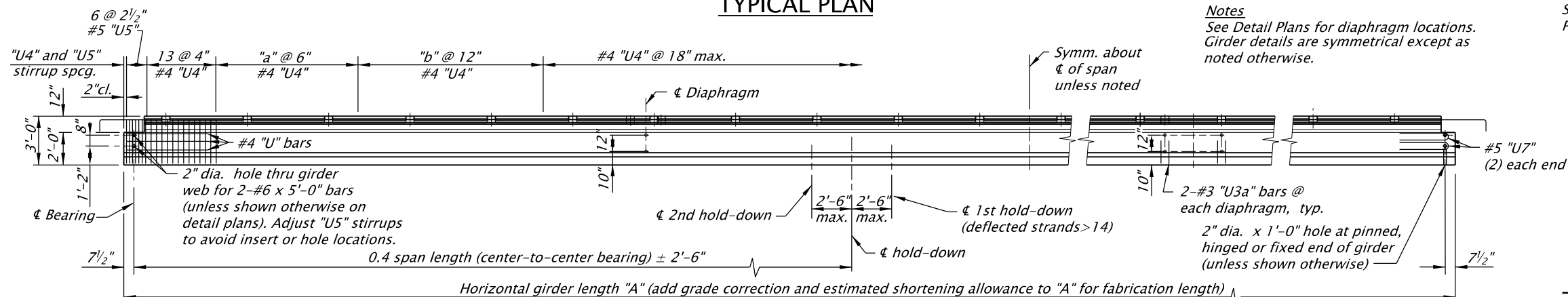
CALC. BOOK NO. - - - -	N/A - - - -	SDR DATE - 20-APR-2018	BR350
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TYPICAL PLAN

Notes

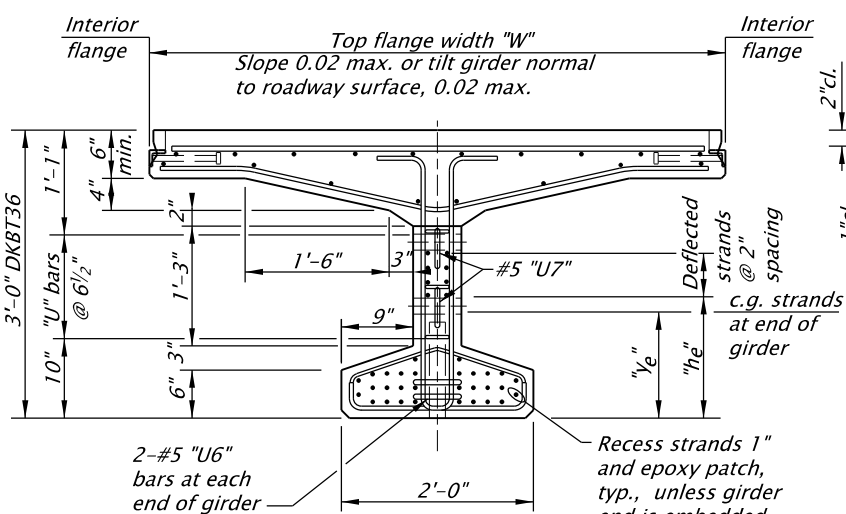
See Detail Plans for diaphragm locations. Girder details are symmetrical except as noted otherwise.



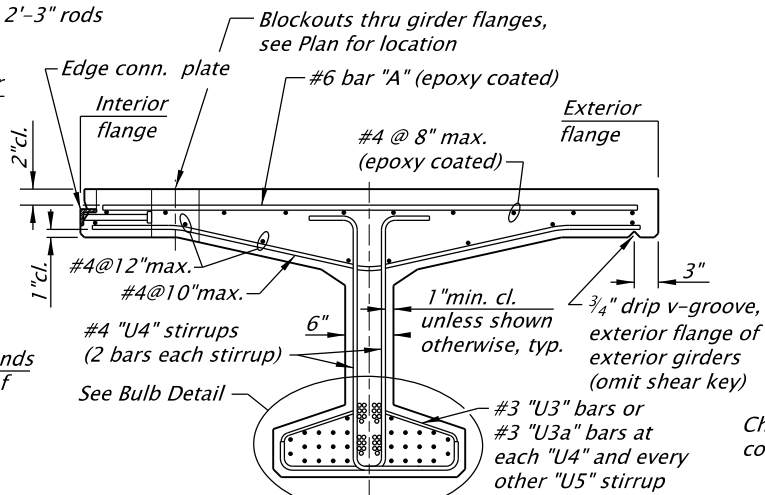
TYPICAL ELEVATION

*CONCRETE INSERTS

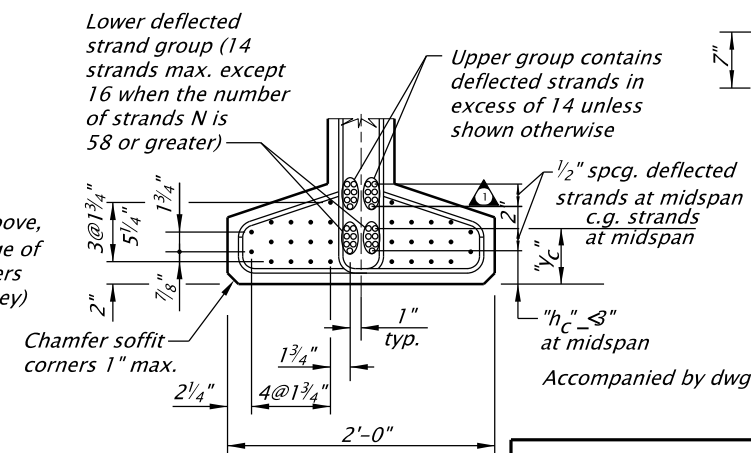
Expanded coil concrete inserts with closed-back ferrule threaded to receive 1" dia. x 2'-3" rods (ASTM A307) threaded full length.



END VIEW



MIDSPAN SECTION



BULB DETAIL

Notes

Girders are designed for live loading and superimposed dead load as shown in the General Notes. Provide the class of concrete shown on the Girder Schedule. Provide concrete strength at transfer of prestress not less than that shown on the Girder Schedule.

Provide 1/2" dia. 7-wire low relaxation prestressing strands conforming to AASHTO M203 (ASTM A416) Grade 270, Supplement 1.

Tension straight strands initially to 31.0 kips. Ensure deflected strands have the same tension after deflecting. Provide reinforcing steel as shown in the General Notes, except as noted.

Maintain girders in an upright position at all times with the support points within 2'-0" of the girder ends.

Provide a transverse broom finish on tops of girders.

Provide general finish for all other surfaces, except as shown in the General Notes.

Deflected strands may be bundled between hold-down points.

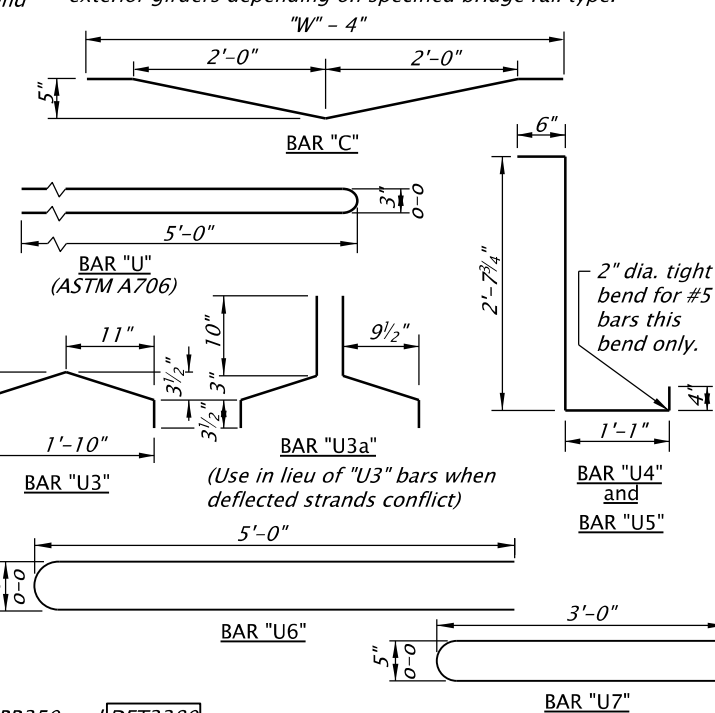
Provide temporary diaphragm beams as required, see BR350.

See dwg. DET3380 for "a", "b", "d", "e", "f" and "theta" values.

Provide welded wire fabric conforming to ASTM A497.

BAR	TOP FLANGE WIDTH "W" (in)		
	60 to 72	72 to 84	84 to 102
"A" Spacing (in)	7	6	5

Additional top deck reinforcement may be required for exterior girders depending on specified bridge rail type.



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OREGON STANDARD DRAWINGS

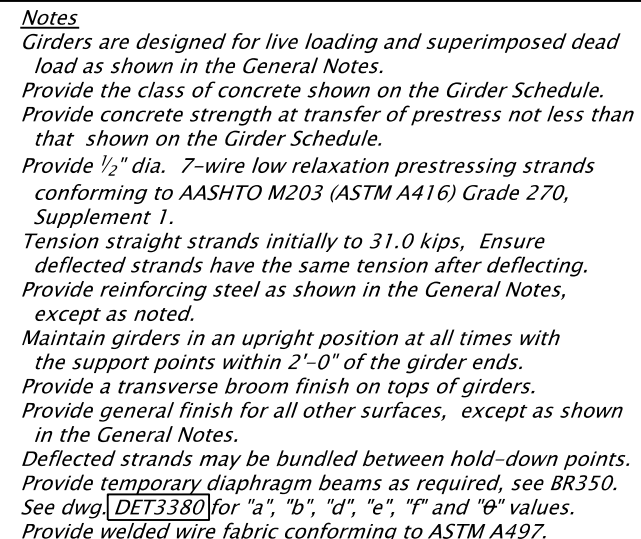
36" DECK BULB-T (PRECAST PRESTRESSED CONCRETE) GIRDER

2024

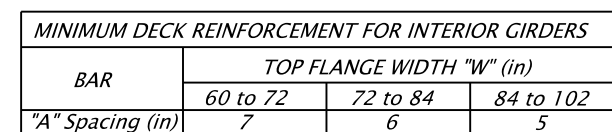
DATE	REVISION	DESCRIPTION
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CALC. BOOK NO. - - -	N/A - - -	SDR DATE - 20-APR-2018 -	BR360
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Effective Date: December 1, 2023 - May 31, 2024

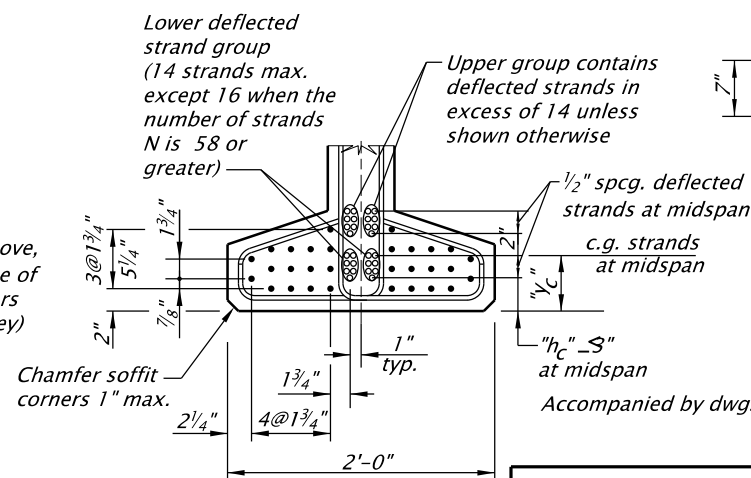


TYPICAL PLAN

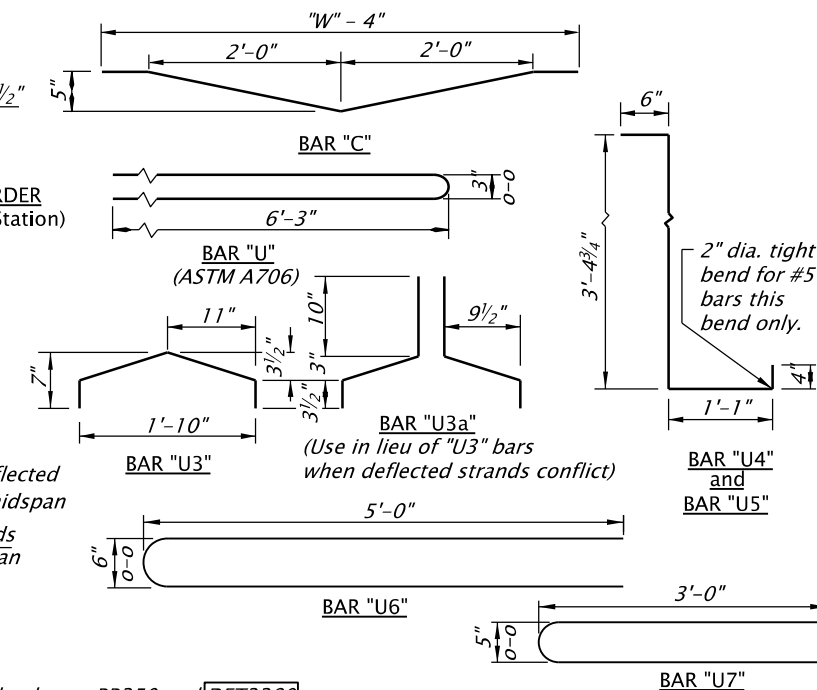


Additional top deck reinforcement may be required for exterior girders depending on specified bridge rail type.

TYPICAL ELEVATION



BULB DETAIL



END VIEW

MIDSPAN SECTION

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All materials shall be in accordance with the current Oregon Standard Specifications.

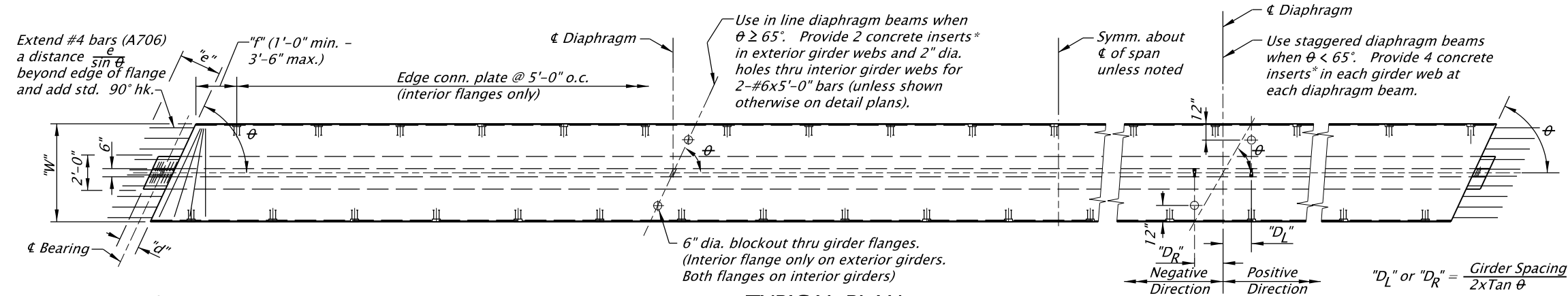
OREGON STANDARD DRAWINGS

**45" DECK BULB-T
(PRECAST PRESTRESSED
CONCRETE) GIRDER**

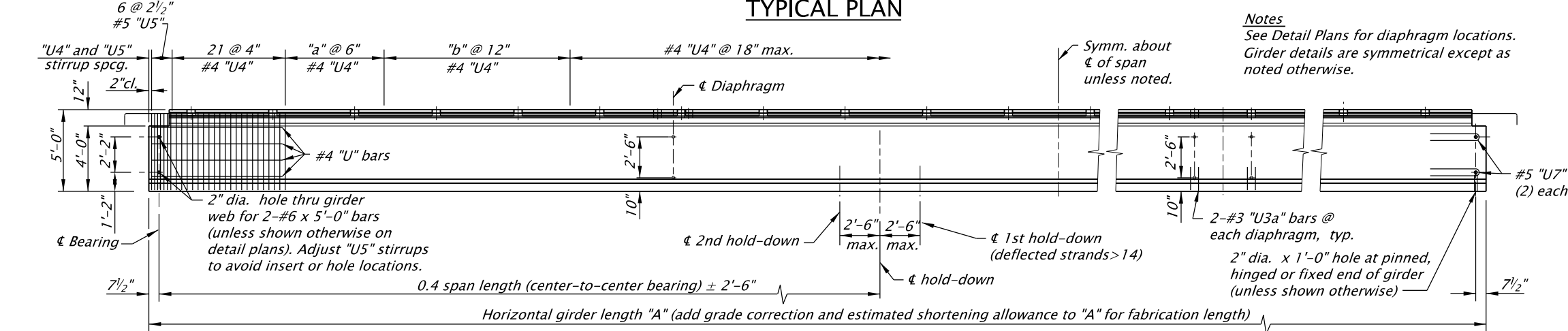
2024

DATE		REVISION DESCRIPTION	
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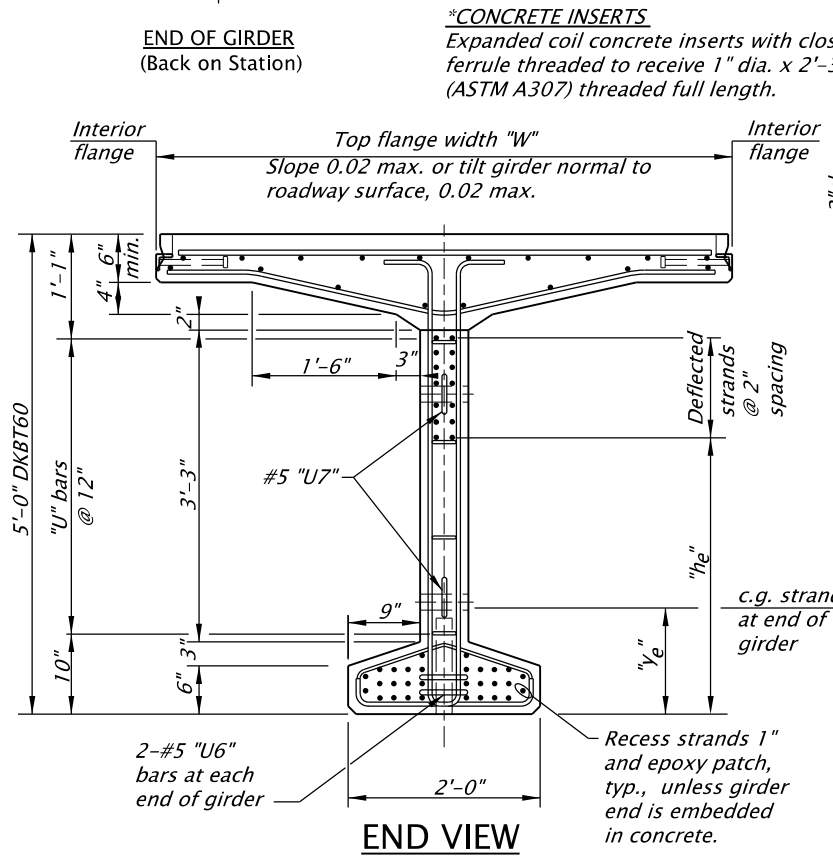
CALC. BOOK NO. - - - -	N/A - - - -	SDR DATE - 20-APR-2018	BR365
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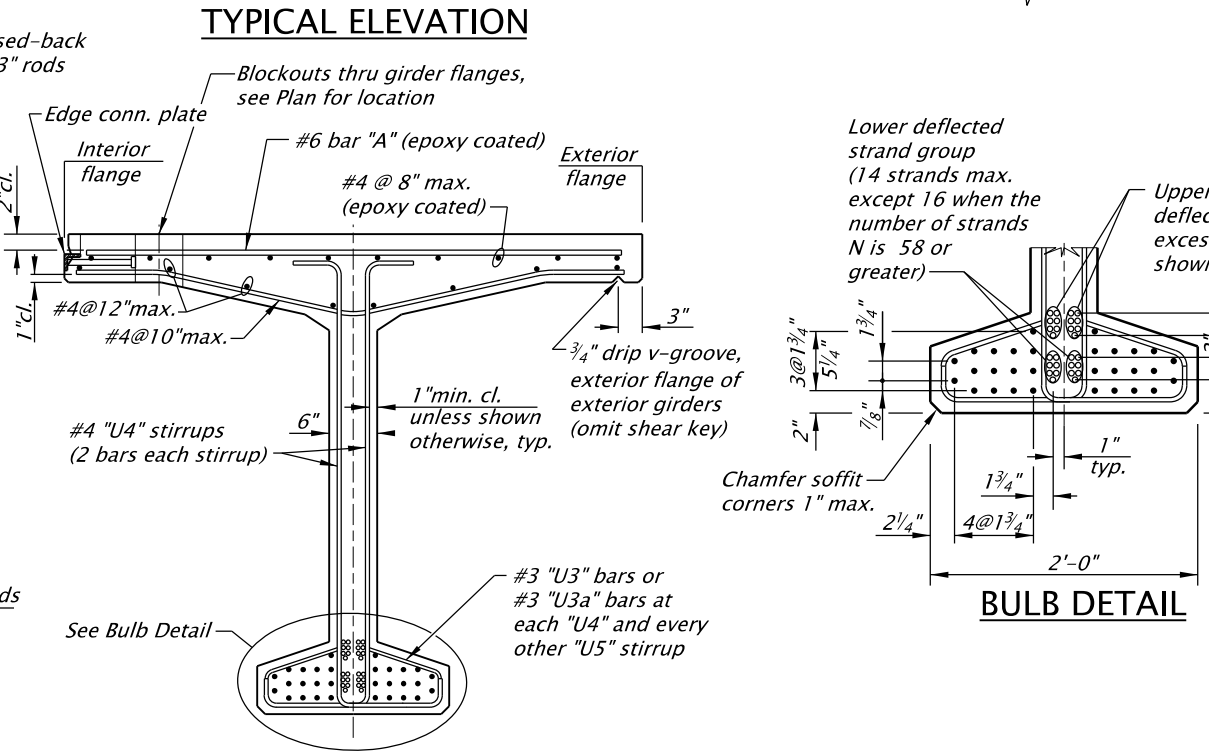
TYPICAL PLAN



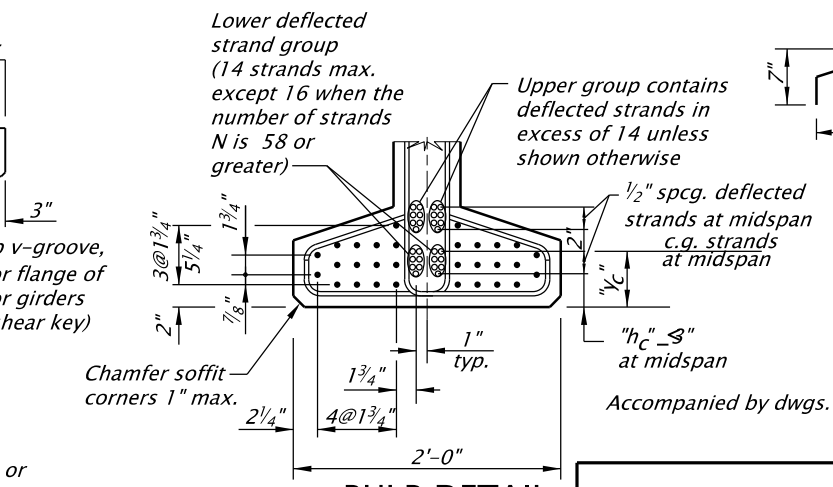
TYPICAL ELEVATION



END VIEW



MIDSPAN SECTION

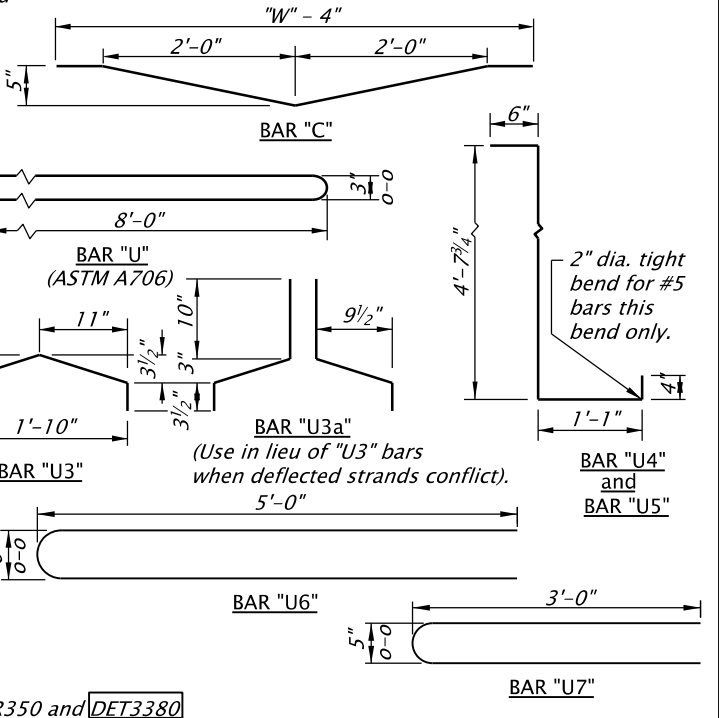


BULB DETAIL

Notes
Girders are designed for live loading and superimposed dead load as shown in the General Notes.
Provide the class of concrete shown on the Girder Schedule.
Provide concrete strength at transfer of prestress not less than that shown on the Girder Schedule.
Provide 1/2" dia. 7-wire low relaxation prestressing strands conforming to AASHTO M203 (ASTM A416) Grade 270, Supplement 1.
Tension straight strands initially to 31.0 kips. Ensure deflected strands have the same tension after deflecting.
Provide reinforcing steel as shown in the General Notes, except as noted.
Maintain girders in an upright position at all times with the support points within 2'-0" of the girder ends.
Provide a transverse broom finish on tops of girders.
Provide general finish for all other surfaces, except as shown in the General Notes.
Deflected strands may be bundled between hold-down points. Provide temporary diaphragm beams as required, see BR350.
See dwg. DET3380 for "a", "b", "d", "e", "f" and "theta" values.
Provide welded wire fabric conforming to ASTM A497.

MINIMUM DECK REINFORCEMENT FOR INTERIOR GIRDERS			
BAR	TOP FLANGE WIDTH "W" (in)		
	60 to 72	72 to 84	84 to 102
"A" Spacing (in)	7	6	5

Additional top deck reinforcement may be required for exterior girders depending on specified bridge rail type.



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OREGON STANDARD DRAWINGS			
60" DECK BULB-T (PRECAST PRESTRESSED CONCRETE) GIRDER			
2024			
DATE	REVISION DESCRIPTION		
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CALC. BOOK NO.	N/A	SDR DATE	20-APR-2018
BR375			